

Medicaid Expansion and Employer-Sponsored Insurance Premiums: Evidence from Double/Debiased Machine Learning

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This paper estimates the spillover effects of Medicaid expansion on employer-sponsored health insurance premiums using Double/Debiased Machine Learning. Exploiting plan-level U.S. Department of Labor Form 5500 filings aggregated to a balanced state-year panel of 1,173 observations across 51 states over 2001–2024, we distinguish between experience-rated contracts (premiums reflecting local claims costs) and non-experience-rated contracts (premiums set across broad actuarial pools). Our primary DML specification includes state fixed effects in the covariate vector, restricting identification to within-state variation; it attenuates experience-rated estimates across all learners, with point estimates ranging from 0.081 to 0.217 (8–20%) depending on learner choice. We find no significant effect on non-experience-rated premiums from the flexible learners, consistent with a cost-shifting mechanism. Removing state fixed effects inflates the experience-rated estimates to 0.212 to 0.300 (19–26%), indicating that between-state differences in premium levels drive much of the pooled result. A standard two-way fixed effects specification produces a significant result for non-experience-rated premiums that DML’s flexible covariate adjustment eliminates, regardless of whether state fixed effects are included. Event study diagnostics reject parallel pre-trends for experience-rated premiums, and a placebo test detects significant effects under a fake 2008 expansion, confirming pre-existing convergence between expansion and non-expansion states. This paper contributes the first DML application to this question, a built-in falsification test using the experience-rated versus non-experience-rated distinction, and a practical demonstration that specification choices in both DML and TWFE substantively change causal estimates in applied settings.

I. Introduction and Literature Review

In 2010 the Affordable Care Act (ACA) was enacted. The ACA gave individual

* An earlier version of this analysis using difference-in-differences was developed jointly with Ameesh Perchani for ECON 490 at UBC. The current paper substantially extends that work with new data, methodology, and analysis; all errors are my own.

states the option to expand Medicaid eligibility to nearly all adults with incomes below 138% of the federal poverty level; primarily childless low-income adults who were mostly ineligible. The Supreme Court’s 2012 NFIB v. Sebelius (Supreme Court of the United States 2012) decision made expansion optional for states. As of 2023, 40 States and the District of Columbia have expanded, while 10 have not. Adoption is staggered from 2014 through 2023; this variation in time and adoption creates a quasi-experimental setting.

Much of the existing Medicaid expansion literature focuses on the directly affected population; newly eligible low-income adults below the 138% Federal Poverty Limit. With such significant newly enacted changes to healthcare and insurance, there may be indirect effects on employer-sponsored insurance (ESI) which covers over 150 million Americans. Previously (Abraham, Royalty and Drake 2019) examine whether Medicaid expansion affected ESI offers, eligibility, take-up, and employee premiums, finding largely null effects. This paper will explore if Medicaid expansion had spillover effects on ESI Premiums. This is policy-relevant as if expansion reduces employer premiums, the benefits extend far beyond the Medicaid-eligible population, and cost-benefit analyses that ignore this channel understate the return on expansion.

The proposed mechanism for this paper is cost-shifting. United States hospitals under the Emergency Medical Treatment and Active Labor Act (EMTALA) are required to treat emergency patients regardless of their ability to pay. When patients are uninsured, hospitals absorb these costs as uncompensated care or shift them to private payers through higher charges. (Garthwaite, Gross and Notowidigdo 2018) finds that "[each] additional uninsured person costs hospitals approximately \$800 each year" The ACA and Medicaid expansion has reduced the uninsured population leading to uncompensated care to fall. (Dranove, Garthwaite and Ody 2016) finds uncompensated care fell from 4.1% to 3.1% of operating costs in expansion states implying that hospitals will have less reason to shift costs as more Americans are insured. This behavior should reduce premiums for employer-sponsored plans specifically the experience-rated plans and not non-experience-rated plans. experience-rated plans set premiums based on the actual claims history of that specific employer’s workforce, so they’re sensitive to local healthcare cost conditions. non-experience-rated plans are priced by insurers across broad risk pools, so local cost changes wash out. This gives us a built-in falsification framework. If our effect is present for both experience and non-experience-rated plans confounding is a more likely explanation than cost-shifting.

The current research landscape leaves a gap in Medicaid’s expansion on its effect on private premiums. From (Dranove, Garthwaite and Ody 2016) we know that expansion reduced uncompensated care costs at hospitals. (Abraham, Roy-

alty and Drake 2019) finds that Medicaid Expansion had minimal effect on ESI offers and take-up rates. To our knowledge, no existing study examines the effect of expansion on ESI premium levels using plan-level filing data. We use plan-level Form 5500 filings from the U.S. Department of Labor, which provide administrative data on premiums distinguishing between experience-rated and non-experience-rated contracts.

Standard DiD approaches are unreliable in this setting for two main reasons: parallel trends are empirically questionable (Section 4 presents event study evidence showing that expansion states’ experience-rated premiums were already converging toward non-expansion states before 2014), and TWFE with staggered treatment can produce biased estimates under heterogeneous effects (Goodman-Bacon 2021). Expansion states are systematically richer, have lower uninsurance rates, and spend more on healthcare than non-expansion states, motivating a method that can flexibly adjust for these differences.

We address these challenges using Double/Debiased Machine Learning (DML) (Chernozhukov et al. 2018). DML replaces parallel trends with conditional unconfoundedness: after controlling for observed state characteristics, expansion status is independent of potential premium outcomes. DML uses machine learning to estimate the relationship between confounders and both the outcome and treatment, then recovers the causal parameter through a residual-on-residual regression. Cross-fitting and Neyman orthogonality ensure that estimation error in the ML models does not contaminate inference on the treatment effect. Our primary specification includes state dummy variables in the covariate vector, restricting identification to within-state variation around the expansion date. The same source of variation as TWFE, but with flexible nonlinear covariate adjustment.

Our primary DML specification attenuates experience-rated estimates across all learners relative to the pooled results, with point estimates ranging from 0.081 to 0.217 depending on learner choice. We find no significant effect on non-experience-rated premiums from the two flexible learners (Random Forest and Gradient Boosting). The null result for non-experience-rated premiums is consistent with our cost-shifting hypothesis and holds across every specification we consider. Removing state fixed effects from the DML covariate vector inflates the experience-rated estimate to 18-26%, revealing that between-state differences in premium levels, not within-state changes around expansion, drive the larger estimates. We further demonstrate that TWFE produces a statistically significant result for non-experience-rated premiums that DML eliminates, regardless of whether state effects are included. This paper thus contributes in three ways: the first DML application to this question, a built-in falsification test using the experience-rated versus non-experience-rated distinction, and a practical demon-

stration that specification choices in both DML and TWFE substantively change causal conclusions.

The paper proceeds as follows. Section 2 describes our data sources: the Form 5500 data, the staggered Medicaid expansion treatment variable, and state-year covariates used in our DML estimation. Section 3 presents our TWFE DiD estimation as a baseline. Section 4 examines pre-trends through event studies in event time, motivating the inclusion of state fixed effects in the DML specification. Section 5 explains our DML framework and presents the main findings. Section 6 presents robustness and mechanism analyses, and Section 7 concludes.

II. Data

In this section we will discuss all of our data used and variable construction.

A. Form 5500 Plan-Level Data

Our main data is the Form 5500 Datasets (U.S. Department of Labor, Employee Benefits Security Administration 2025), from the U.S. Department of Labor. It is plan level data for every employer-sponsored benefit plan in the United States. We cover 2001-2024 excluding 2004 because of major outliers and data quality issues. We use Schedule A which contains insurance information, specifically the variables: `WLFR_PREMIUM_RCVD_AMT`, premiums received for experience-rated contracts `WLFR_TOT_CHARGES_PAID_AMT`, total charges paid for non-experience-rated contracts `INS_PRSN_COVERED_EOY_CNT`, persons covered at end of year. experience-rated contracts are arrangements in which the employer bears the underwriting risk and premiums reflect the plan’s own claims history. non-experience-rated contracts are conventional insured arrangements in which the carrier bears risk and sets premiums based on broader actuarial pools. The distinction between experience-rated and non-experience-rated is built into the Schedule A structure. From the main Form 5500 we use `SPONS_DFE_MAIL_US_STATE` as the state identification variable for 2009 onwards, and `SPONS_DFE_STATE` for earlier years.

From these variables we construct two outcome variables of per-capita premiums (total premiums divided by persons covered at end of year), one for experience-rated plans and one for non-experience-rated plans. Both of these outcome variables are log-transformed; mean log premium per capita at the plan-year level. This is done in order to reduce skewness and allow percentage interpretation of coefficients.

For data cleaning we remove the 5th and 95th percentile outliers on per-capita premiums, years 1999,2000 are dropped due to sparse coverage and low data volume. 2004 is dropped due to outliers and data validity concerns. Along with

this Observations with zero or negative persons covered are excluded. This leaves us with a balanced panel with 1,173 observations at the state-year level (51 states x 23 years).

Table 1 presents summary statistics for the full sample and by expansion status. Never-expansion states have notably higher mean experience-rated premiums (2,332 $vs.$ 1,677), consistent with the cost-shifting hypothesis these states have more uninsured residents generating uncompensated care costs that may be passed to private payers. The DML analysis accounts for these and other level differences through flexible covariate adjustment.

Table 1—: Summary Statistics: State-Year Level, 2001–2024

	N	Mean	SD	Min	Median	Max
<i>Panel A: Full Sample</i>						
Log Premium (Exp-Rated)	1173	6.548	0.464	5.149	6.507	7.909
Premium (Exp-Rated, \$)	1173	1,805	860	256	1,628	5,393
Plans per State-Year (Exp-Rated)	1173	435	468	5	305	3,370
Log Premium (Non-Exp-Rated)	1173	5.963	0.244	5.113	5.953	6.864
Premium (Non-Exp-Rated, \$)	1173	1,074	323	349	1,047	2,676
Plans per State-Year (Non-Exp-Rated)	1173	3,687	4,599	83	2,165	31,379
<i>Panel B: Expansion States (41)</i>						
Log Premium (Exp-Rated)	943	6.468	0.405	5.534	6.427	7.793
Premium (Exp-Rated, \$)	943	1,677	743	370	1,549	5,064
Plans per State-Year (Exp-Rated)	943	422	494	6	255	3,370
Log Premium (Non-Exp-Rated)	943	5.995	0.244	5.284	5.975	6.864
Premium (Non-Exp-Rated, \$)	943	1,124	320	349	1,075	2,676
Plans per State-Year (Non-Exp-Rated)	943	3,598	4,738	137	1,992	31,379
<i>Panel C: Never-Expansion States (10)</i>						
Log Premium (Exp-Rated)	230	6.878	0.540	5.149	6.789	7.909
Premium (Exp-Rated, \$)	230	2,332	1,082	256	1,993	5,393
Plans per State-Year (Exp-Rated)	230	489	337	5	404	1,477
Log Premium (Non-Exp-Rated)	230	5.833	0.201	5.113	5.813	6.263
Premium (Non-Exp-Rated, \$)	230	866	243	406	871	1,469
Plans per State-Year (Non-Exp-Rated)	230	4,054	3,967	83	2,422	16,788

Notes: Unit of observation is the state-year. Premiums are per-capita annual amounts in nominal dollars, computed from Form 5500 Schedule A filings. experience-rated plans report premiums received; non-experience-rated plans report total charges paid. Plan-level observations are trimmed at the 5th and 95th percentiles before aggregation. Data cover 51 states (including DC) over 2001–2024, excluding 2004. Expansion states are the 41 states (including DC) that expanded Medicaid between 2014 and 2023; never-expansion states are AL, FL, GA, KS, MS, SC, TN, TX, WI, and WY.

Dollar premiums reflect arithmetic means; log premiums reflect the mean of log transformed values across plans within each state-year.

B. Treatment Variable: Staggered Medicaid Expansion

Our treatment variable is: D_{st} a binary indicator which represents if a state has expanded Medicaid by year t , zero otherwise. This is constructed from (Kaiser Family Foundation 2025). Medicaid Expansion is staggered with 27 states in January of 2014, then smaller waves 2015 (3 states), 2016 (2 states), 2019 (2 states), 2020 (3 states), 2021 (2 states) and 2023 (2 states). Ten states have never expanded. One key outlier is Wisconsin which is classified as non-expansion despite covering adults up to 100% FPL, because it did not adopt the full ACA expansion to 138% FPL and does not receive the enhanced federal matching rate.

Table 2—: Medicaid Expansion Cohorts

Cohort Year	States	N
2014	AR, AZ, CA, CO, CT, DC, DE, HI, IA, IL, KY, MA, MD, MI, MN, NH, NJ, NM, NV, NY, ND, OH, OR, RI, VT, WA, WV	27
2015	AK, IN, PA	3
2016	LA, MT	2
2019	ME, VA	2
2020	ID, NE, UT	3
2021	MO, OK	2
2023	NC, SD	2
Never	AL, FL, GA, KS, MS, SC, TN, TX, WI, WY	10
Total		51

Notes: Expansion year is the calendar year in which the state’s Medicaid expansion took effect. For states expanding mid-year, we assign the calendar year of implementation since our data is at the annual level. Sources: Congressional Research Service (IF10399) (Congressional Research Service 2021), Kaiser Family Foundation, advisory.com. Wisconsin covers adults up to 100% FPL but did not adopt the full ACA expansion and is classified as non-expansion.

C. State-Year Covariates

We merge eleven state-year covariates onto the panel to serve as the confounder set X in the DML specification. Table 3 summarizes each variable, its source, and coverage.

Covariate availability varies by source. Healthcare spending per capita is available only through 2019 (CMS State Health Expenditure Accounts), and uninsurance rates begin in 2009 (Census ACS), restricting the complete-covariate DML sample to 550 state-year observations over 2009–2019 across 50 states. The full panel (1,173 observations) is used for TWFE estimation. Dropping healthcare spending per capita from the covariate set would extend the DML sample to approximately 750 observations through 2024; we note this tradeoff but retain the variable in our primary specification given the direct relevance of healthcare supply costs to the cost-shifting mechanism.

Table 3—: State-Year Covariates Used in DML Estimation

Variable	Source	Years
<i>Labor Market</i>		
Unemployment Rate (%)	BLS Local Area Unemployment Statistics	2001–2024
<i>Income and Poverty</i>		
Median Household Income (\$)	Census ACS (Table B19013)	2005–2024
Poverty Rate (%)	Census ACS (Table S1701)	2009–2024
<i>Demographics</i>		
Population	Census ACS (Table B01003)	2005–2024
Share Aged 65+ (%)	Census ACS (Table S0101)	2005–2024
Share Non-Hispanic White (%)	Census ACS (Table B03002)	2005–2024
<i>Insurance Market</i>		
Uninsurance Rate (%)	Census ACS (Table S2701)	2009–2024
State-Based Marketplace	KFF / hardcoded	Time-invariant
Pre-ACA Guaranteed Issue Laws	NAIC / hardcoded	Time-invariant
<i>Healthcare Supply</i>		
Healthcare Spending per Capita (\$)	CMS State Health Expenditure Accounts	2005–2019

Notes: All covariates are at the state-year level. Year fixed effects (22 dummy variables, one omitted) are also included in the DML covariate set, bringing the total number of covariates to 32. Time-invariant indicators do not vary across years within a state. Sources: BLS Local Area Unemployment Statistics (U.S. Bureau of Labor Statistics 2025); U.S. Census Bureau American Community Survey (U.S. Census Bureau 2025); CMS State Health Expenditure Accounts (Centers for Medicare and Medicaid Services 2023); Kaiser Family Foundation (Kaiser Family Foundation 2025).

D. Sample Description

We use three samples. The **full sample** (N=1,173) consists of all state-years 2001–2024 with premium data and is used for TWFE DiD estimation. The **DML sample** (N=550) restricts to state-years 2009–2019 with complete covariates across all sources (see Section 2.3) and is used for DML estimation and the apples-to-apples TWFE comparison. The **pre-expansion sample** (N=250) further restricts to 2009–2013 and is used for placebo tests.

E. Covariate Balance

Expansion and non-expansion states differ systematically on pre-treatment characteristics. This motivates the use of DML over standard DiD. Standardized differences exceeding 0.25 on income (0.93), poverty (0.75), uninsurance (0.79), and healthcare spending (0.60) indicate that expansion and non-expansion states differ substantially on characteristics that plausibly affect premium levels. Linear regression adjustments may be insufficient for differences like this which interact with premiums.

Table 4—: Pre-2014 Covariate Balance: Expansion vs. Never-Expansion States

	Expansion (41)		Never-Expansion (10)		Std.
	Mean	(SD)	Mean	(SD)	Diff.
Unemployment Rate (%)	6.08	(2.15)	6.47	(2.07)	−0.18
Median Household Income (\$)	51,844	(8,477)	45,318	(5,178)	0.93
Poverty Rate (%)	13.96	(3.07)	16.41	(3.44)	−0.75
Uninsurance Rate (%)	13.06	(4.03)	16.21	(3.92)	−0.79
Population 65+ (%)	13.16	(1.71)	13.10	(1.98)	0.03
Non-Hispanic White (%)	71.76	(16.55)	67.93	(12.70)	0.26
Healthcare Spending/Capita (\$)	7,119	(1,285)	6,497	(715)	0.60

Notes: Means and standard deviations computed over state-year observations for 2009–2013 (years with complete covariate coverage). Standardized difference is $(\bar{x}_{\text{exp}} - \bar{x}_{\text{non}})/\sqrt{(s_{\text{exp}}^2 + s_{\text{non}}^2)/2}$. Values exceeding |0.25| indicate meaningful imbalance. Sources: Census ACS (income, demographics, poverty, uninsurance), BLS LAUS (unemployment), CMS State Health Expenditure Accounts (healthcare spending).

Figure 1 plots premium trends by expansion cohort. Panel A shows that never expansion states (red) have consistently higher experience-rated premiums across the full period. Panel B, which aligns all cohorts in event time, reveals a declining pre-trend in experience-rated premiums prior to expansion, motivating the event study analysis and DML approach in subsequent sections.

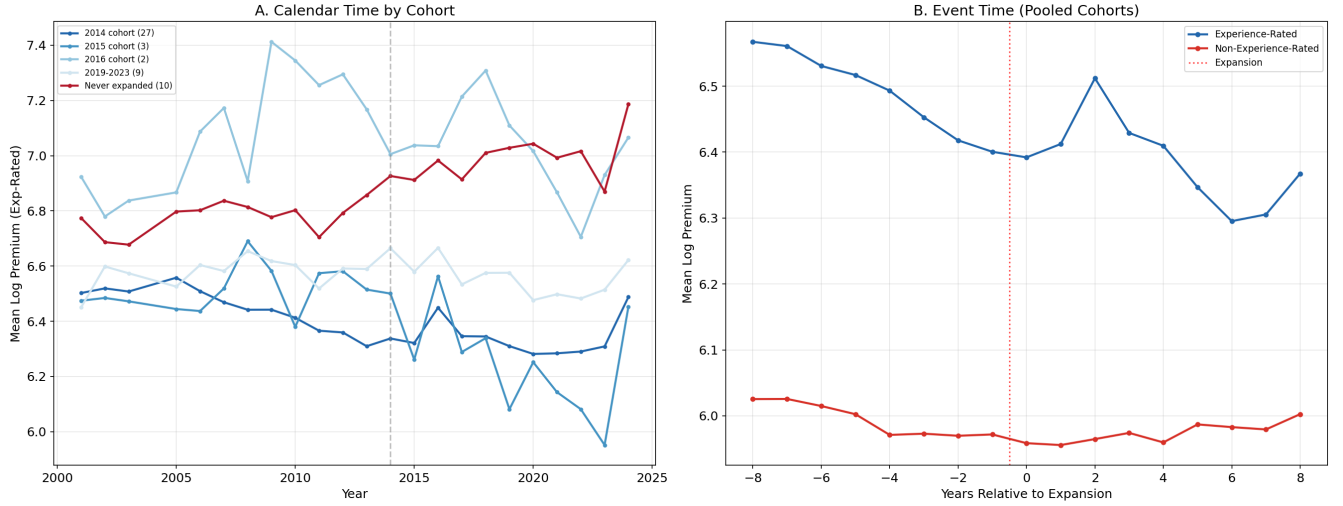


Figure 1. : Premium Trends Around Medicaid Expansion

Notes: Panel A plots mean log per-capita experience-rated premiums by expansion cohort in calendar time. Panel B plots mean log premiums in event time (years relative to each state’s expansion date), averaged across all expansion cohorts. Premiums are computed from Form 5500 Schedule A filings, trimmed at the 5th and 95th percentiles at the plan level before aggregation to the state-year level.

III. TWFE DiD

For our baseline specification we employ a standard two-way fixed effect (TWFE) model:

$$(1) \quad \log(\text{Premium}_{st}) = \alpha_s + \gamma_t + \theta D_{st} + \varepsilon_{st}$$

where α_s are state fixed effects, γ_t are year fixed effects, and D_{st} is our treatment variable of staggered Medicaid Expansion, representing if state s has implemented Medicaid expansion by year t . We cluster standard errors at the state level to account for serial correlation within states. We note that TWFE with staggered treatment can produce biased estimates under heterogeneous effects (Goodman-Bacon 2021). Under heterogeneous treatment effects across expansion states, the TWFE estimator is a weighted average of all possible two-by-two DiD comparison. There is a possibility of negative weights for later-treated states. Along with parallel trends this motivates our choice of DML as an alternative main specification as DML does not impose treatment effect homogeneously.

Our results are shown in Table 5:

In our full sample we find a 17.8% reduction in experience-rated premiums

Table 5—: TWFE DiD Estimates: Effect of Medicaid Expansion on Premiums

	Full Sample		DML-Comparable Sample	
	Exp-Rated	Non-Exp-Rated	Exp-Rated	Non-Exp-Rated
$\hat{\theta}$	−0.196** (0.078)	−0.086** (0.038)	−0.118** (0.059)	−0.047** (0.022)
95% CI	[−0.349, −0.043]	[−0.161, −0.010]	[−0.232, −0.003]	[−0.089, −0.005]
p -value	0.012	0.026	0.044	0.029
R^2	0.732	0.735	0.877	0.913
N	1,173	1,173	550	550
Clusters	51	51	50	50
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: Dependent variable is mean log per-capita premium at the state-year level. experience-rated premiums reflect premiums received by insurers; non-experience-rated reflect total charges paid. Standard errors clustered at the state level in parentheses. The full sample covers 51 states over 2001–2024 (excluding 2004). The DML-comparable sample covers 50 states and restricts to 2009–2019 with complete covariate coverage. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

($p=0.012$) and a 8.2% reduction in non-experience-rated premiums ($p=0.026$). For the DML-Comparable sample we restrict our coverage to 50 states from 2009–2019 (for complete covariate coverage). We find a 11.1% reduction in experience-rated premiums ($p=0.044$) and a 4.6% reduction in non-experience-rated premiums ($p=0.029$). In both of our samples we have a significant decrease in premiums with this effect being greater in experience rated plans. The coefficients change substantially between the Full Sample and DML-Comparable Sample (−0.196 to −0.118 and −0.086 to −0.047). The longer pre-period and post period contribute to these larger coefficients. We report both, but focus more on the DML-Comparable Sample to allow us to have an apples-to-apples comparison for our DML specification. Before continuing on to our DML estimation, we examine the parallel trends assumption underlying TWFE through an event study analysis.

IV. Event Study

We estimate a dynamic event study in event time (years are calculated relative to each state’s Medicaid expansion years) not calendar years. Each state’s coefficients are measured relative to its own expansion year.

$$(2) \quad \log(\text{Premium}_{st}) = \alpha_s + \gamma_t + \sum_{\substack{e=-5 \\ e \neq -1}}^5 \delta_e \cdot \mathbf{1}[\text{EventTime}_{st} = e] + \varepsilon_{st}$$

where $\text{EventTime}_{st} = t - E_s$ is the number of years relative to state s ’s expansion year E_s , and $e = -1$ is the omitted reference period. We bin endpoints at $e \leq -5$

and $e \geq 5$ to avoid thin cells. Only expansion states enter the regression; never-expanded states contribute implicitly through the year fixed effects γ_t . Standard errors are clustered at the state level.

The pre-trends F-test jointly tests $H_0 : \delta_{-5} = \delta_{-4} = \delta_{-3} = \delta_{-2} = 0$. Rejection in this test means that expansion and non-expansion states were already diverging before treatment, which violates our parallel trends assumption; necessary for TWFE DiD.

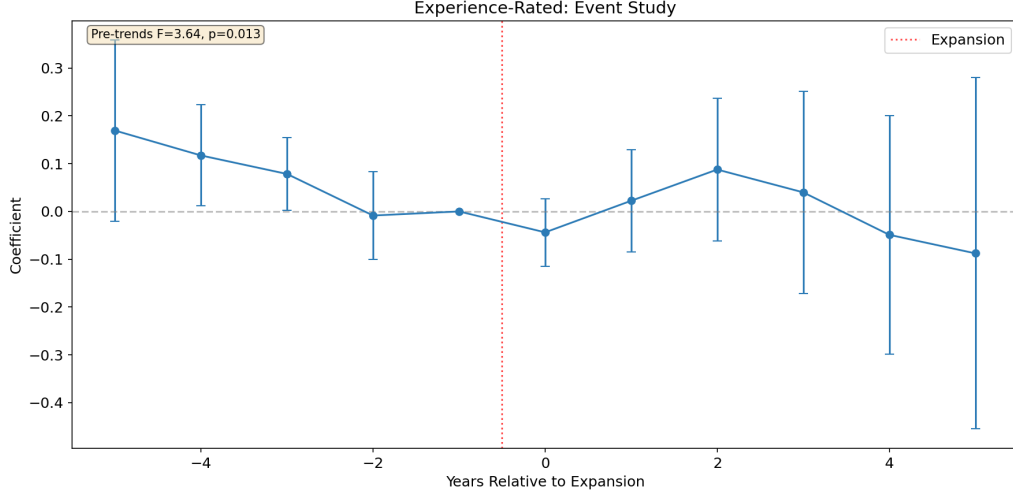


Figure 2. : Event Study: experience-rated Premiums

Notes: Coefficients $\hat{\delta}_e$ from Equation (2) with 95% confidence intervals. Reference period is $e = -1$. Pre-trends joint F-test: $F = 3.64$, $p = 0.013$, rejecting parallel pre-trends at the 5% level. State and year fixed effects included; standard errors clustered at the state level.

Figure 2 shows the event study for experience-rated Plans. The pre-period coefficients are at $e = -5(0.17)$ then $e = -4(0.12)$ and are individually significant. They later decline towards 0 at $e = -2(-0.01)$. This downward trend before expansion indicates that expansion states' experience-rated premiums were already converging toward non-expansion states before Medicaid Expansion began in 2014. The post-expansion coefficients first rise and then fall again, but none are significantly differentiable from zero. This suggests a pre-expansion convergence rather than a clear treatment effect.

Figure 3 does not show the same patterns as we had for experience rated plans; non-experience-rated premiums have pre-period coefficients which are small and statistically insignificant. The F-test cannot reject parallel trends.

Parallel trends do not pass for experience-rated plans ($F=3.64$, $p = 0.013$), but pass for non-experience-rated ($F = 1.24$, $p = 0.309$). This finding means

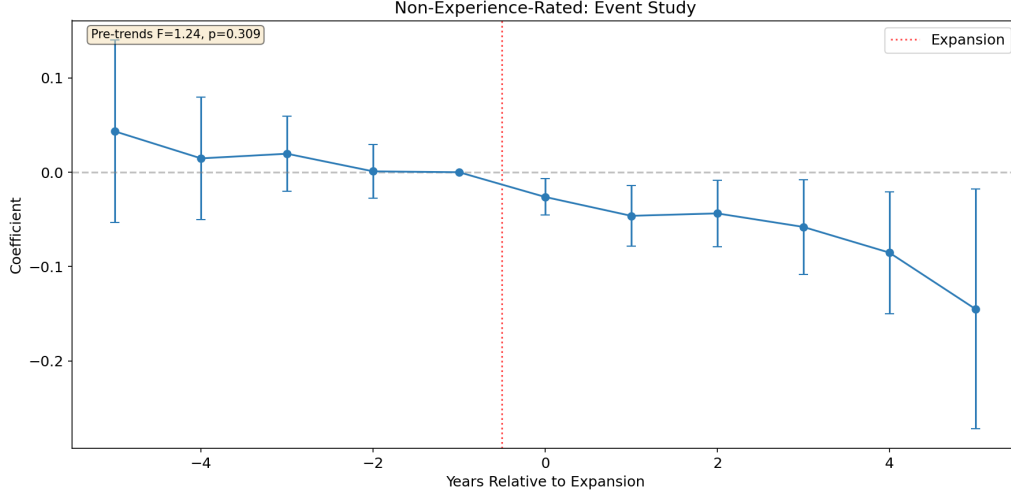


Figure 3. : Event Study: non-experience-rated Premiums

Notes: Coefficients $\hat{\delta}_e$ from Equation (2) with 95% confidence intervals. Reference period is $e = -1$. Pre-trends joint F-test: $F = 1.24$, $p = 0.309$, failing to reject parallel pre-trends. State and year fixed effects included; standard errors clustered at the state level.

that our TWFE DiD estimate for experience-rated premiums is unreliable. The estimate in Table 5 may be biased by pre-existing differential trends. This pre-trends pattern is a between-state phenomenon: expansion states were converging toward non-expansion states in experience-rated premium levels before treatment. This motivates including state dummy variables in the DML covariate vector to absorb these persistent level differences, restricting identification to within-state variation. For non-experience-rated plans (where parallel trends hold) the estimates from TWFE and DML should agree; which we verify in Section 5.

We now continue with the DML framework, which addresses the identification concerns raised by the event study.

V. DML

For our main specification we estimate the Partially Linear Regression (PLR) model from (Chernozhukov et al. 2018). Using the DoubleML Python Package (Bach et al. 2022).

$$(3) \quad Y_{st} = \theta D_{st} + g(X_{st}) + \varepsilon_{st}$$

where Y_{st} is the mean log per-capita premium in state s at time t , D_{st} is the staggered Medicaid expansion indicator, X_{st} is a vector of 32 state-year covariates

(including year fixed effects), and $g(\cdot)$ is an unknown nuisance function. The parameter of interest is θ , the causal effect of expansion on premiums after flexibly controlling for confounders.

DML estimates θ using partialling-out estimator:

$$(4) \quad \hat{\theta}_{DML} = \left(\frac{1}{N} \sum_{i=1}^N \tilde{D}_i^2 \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \tilde{D}_i \tilde{Y}_i \right)$$

where $\tilde{Y}_i = Y_i - \hat{\ell}(X_i)$ and $\tilde{D}_i = D_i - \hat{m}(X_i)$ are residuals from ML estimates of $E[Y|X]$ and $E[D|X]$, respectively. We use 5-fold cross-fitting with 5 repetitions to avoid overfitting bias. Neyman orthogonality ensures that estimation error in the nuisance functions $\hat{\ell}$ and $\hat{m}$ does not contaminate inference on θ (Chernozhukov et al. 2018).

To ensure robustness of our estimator we employ three learners. Random Forest (Breiman 2001) with 300 trees, and a max depth of 6, Gradient Boosting (Friedman 2001) with 500 iterations at a depth of 4 with a learning rate of 0.05 and finally Lasso (Tibshirani 1996) with cross-validation penalty. Our covariate vector includes unemployment rate, median household income, population, share aged 65+, share non-Hispanic white, poverty rate, uninsurance rate, healthcare spending per capita, marketplace type, pre-ACA guaranteed issue status, year fixed effects, and 49 state dummy variables (approximately 81 variables total). The state dummies restrict identification to within-state variation around the expansion date. The DML sample comprises 550 state-year observations over 2009-2019, restricted by covariate availability.

Our primary specification attenuates the experience-rated estimates across all three learners relative to the pooled results reported in Table 7. Random Forest yields -0.217 ($p < 0.001$), Gradient Boosting yields -0.081 ($p = 0.077$), and Lasso yields -0.140 ($p = 0.003$). The degree of attenuation ranges from 21% (Random Forest) to 62% (Gradient Boosting), confirming that between-state variation inflates the pooled estimates. That the magnitude varies substantially across learners indicates sensitivity to how flexibly the nuisance function handles the expanded covariate vector. Gradient Boosting, which achieves the highest outcome model fit ($R_Y^2 = 0.995$), attenuates the most, suggesting that better modeling of the premium-covariate relationship absorbs more of the between-state confounding. The estimates are directionally consistent with a premium reduction but the precise magnitude—ranging from 8% to 20% depending on learner—remains uncertain.

For non-experience-rated premiums, Random Forest and Gradient Boosting find null effects (0.004 and 0.007, both $p > 0.6$), consistent with the pooled

Table 6—: DML Estimates: Effect of Medicaid Expansion on Premiums (Primary Specification)

Outcome	Learner	$\hat{\theta}$	SE	p	R_Y^2	R_D^2
<i>Panel A: Experience-Rated Premiums</i>						
	Random Forest	−0.217***	0.058	0.000	0.970	0.909
	Gradient Boosting	−0.081*	0.046	0.077	0.995	0.918
	Lasso	−0.140***	0.048	0.003	0.980	0.975
	TWFE DiD	−0.118**	0.059	0.044	—	—
<i>Panel B: Non-Experience-Rated Premiums</i>						
	Random Forest	0.004	0.017	0.809	0.985	0.884
	Gradient Boosting	0.007	0.013	0.601	0.998	0.943
	Lasso	−0.027**	0.011	0.018	0.993	0.970
	TWFE DiD	−0.047**	0.022	0.029	—	—

Notes: DML estimates from the partially linear model (Equation 3) using 5-fold cross-fitting with 5 repetitions. The covariate vector includes 49 state dummy variables (81 total covariates), restricting identification to within-state variation. R_Y^2 and R_D^2 are approximate out-of-sample R^2 for the nuisance models $E[Y|X]$ and $E[D|X]$. TWFE includes state and year fixed effects with clustered standard errors. $N = 550$ state-year observations (2009–2019, 50 states). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

specification and the cost-shifting hypothesis. Lasso yields a small significant estimate (−0.027, $p = 0.018$), likely reflecting its limited flexibility in absorbing the expanded covariate vector; we do not interpret this as evidence against the falsification test. Crucially, TWFE produces a significant −0.047 ($p = 0.029$) for non-experience-rated premiums on the same sample. The two flexible DML learners eliminate this result regardless of whether state effects are included, confirming that the TWFE finding was driven by confounding.

The high nuisance model R^2 values (0.970 – 0.995 for the outcome model, 0.909 – 0.975 for the treatment model) indicate that the ML learners successfully capture the relationship between covariates and both premiums and expansion status. The treatment model R^2 of 0.938 also indicates that expansion status is nearly fully predicted by covariates and state identity, leaving limited residual variation for identification a trade-off inherent in within-state DML with a small number of treatment-timing cohorts.

To assess how much the choice of identifying variation matters, we re-estimate the PLR model without state dummy variables, allowing the treatment effect to be identified from both within-state changes and between-state differences in premium levels. Table 7 reports these pooled estimates alongside the primary within-state results.

Table 7—: DML Estimates: Effect of Medicaid Expansion on Premiums

Outcome	Learner	$\hat{\theta}$	SE	p	R_Y^2	R_D^2
<i>Panel A: experience-rated Premiums</i>						
	Random Forest	−0.286***	0.056	0.000	0.970	0.892
	Gradient Boosting	−0.206***	0.054	0.000	0.985	0.908
	Lasso	−0.301***	0.059	0.000	0.927	0.926
	TWFE DiD	−0.118**	0.059	0.044	—	—
<i>Panel B: non-experience-rated Premiums</i>						
	Random Forest	0.011	0.018	0.535	0.984	0.894
	Gradient Boosting	−0.005	0.016	0.772	0.994	0.916
	Lasso	0.025	0.022	0.246	0.944	0.926
	TWFE DiD	−0.047**	0.022	0.029	—	—

Notes: DML estimates from the partially linear model (Equation 3) using 5-fold cross-fitting with 5 repetitions. The covariate vector includes 49 state dummy variables, restricting identification to within-state variation. [rest of existing notes]. $N = 550$ state-year observations (2009-2019, 50 states). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A. DML without State Fixed Effects

Removing state fixed effects inflates the experience-rated coefficient from 0.081 to 0.212 for Gradient Boosting, from 0.217 to 0.276 for Random Forest, and from 0.140 to 0.300 for Lasso. All pooled estimates are significant at $p < 0.01$. This indicates that the majority of the estimated effect in the pooled specification comes from between-state variation—the persistent difference in premium levels between expansion and non-expansion states—rather than from within-state changes around the expansion date. The non-experience-rated null result holds in both specifications, confirming that the falsification test is robust to the choice of identifying variation. The gap between within-state and pooled estimates is the central finding of this paper: it demonstrates that DML estimates are sensitive to whether state-level heterogeneity is absorbed, a point with general applicability beyond this specific policy question.

VI. Robustness and Mechanisms

A. Pre-trends diagnosis

In order to have a better understanding of our event study (section 4) and reasons for rejecting parallel trends for experience-rated premiums ($F = 3.64$, $p = 0.013$). We inspect individual pre-period coefficients. $\hat{\delta}_e$ for $e \in -5, -4, -3, -2$ (with $e = -1$ as the normalized reference) to find if rejection is a result of either a systemic trend or isolated outlier years.

For experience-rated premiums two of our four pre-period coefficients are indi-

Table 8—: Pre-Period Event Study Coefficients

Event Time	experience-rated			non-experience-rated		
	$\hat{\delta}_e$	SE	Sig.	$\hat{\delta}_e$	SE	Sig.
$e = -5$	0.169	0.097		0.043	0.049	
$e = -4$	0.117	0.054	**	0.015	0.033	
$e = -3$	0.078	0.039	**	0.020	0.020	
$e = -2$	-0.009	0.047		0.001	0.015	
$e = -1$	(reference)			(reference)		
Pre-trend slope	-0.057			-0.014		
F-test	$F = 3.64, p = 0.013$			$F = 1.24, p = 0.309$		

Notes: Coefficients from Equation (2). Reference period is $e = -1$ (year before expansion). Event time $e = -5$ bins all observations five or more years before expansion. Pre-trend slope is from a linear fit of $\hat{\delta}_e$ on e for $e \in \{-5, -4, -3, -2\}$. The F-test jointly tests $H_0 : \delta_{-5} = \delta_{-4} = \delta_{-3} = \delta_{-2} = 0$. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

vidually significant ($e = -4$ and $e = -3$). This pattern points to the fact that Medicaid expansion states had moderately higher premiums several years before expansion (0.078 to 0.169 log points) with the gap narrowing towards zero at $e = -2$. This linear downward pre-trend slope suggests convergence of expansion and non-expansion states before treatment. The F-test rejection is driven by *level* differences in the distant pre-period rather than an accelerating trend that would behave like a treatment effect.

For non-experience-rated premiums, all of our pre-period coefficients are small and statistically insignificant, the F-Test fails to reject parallel trends ($p = 0.309$). This is a very important contrast that suggests the pre-period differences originate from the specific plan type (Experience vs Non-Experience), not some state-level confound.

Our DML estimator does not require parallel trends. Instead DML uses conditional unconfoundedness: treatment assignment is independent of potential outcomes after conditioning on X_{st} . This pre-trend behavior motivates our DML approach. However, if the pre-existing convergence reflects factors which are not captured by our covariate vector our DML estimate may also be affected. We assess this concern through placebo tests in Section 6.

B. Mechanism Test

For our mechanism test we test the first link in our hypothetical causal chain. Medicaid Expansion led to reduced uninsurance which led to less uncompensated care which led to lower cost shifting and lower employer premiums. We ask "Did expansion actually reduce state-level uninsurance rates?"

$$(5) \quad \text{Uninsurance}_{st} = \theta^U D_{st} + g(X_{st}^{-U}) + \varepsilon_{st}$$

where X_{st}^{-U} denotes the covariate vector excluding the uninsurance rate (which is now the outcome). We estimate Equation (5) using both TWFE and DML on the same 550-observation sample. For DML we use Gradient Boosting with the same cross-fitting procedure as the main results.

Table 9—: Mechanism Test: Effect of Medicaid Expansion on State Uninsurance Rate

Method	$\hat{\theta}^U$	SE	p -value
TWFE DiD	−1.268***	0.464	0.006
DML (GBR)	−1.543***	0.240	0.000

Notes: Dependent variable is the state-year uninsurance rate (percent). Both specifications use the 550-observation DML-comparable sample (2009–2019). TWFE includes state and year fixed effects with clustered standard errors. DML uses Gradient Boosting with the same cross-fitting procedure as the main results, excluding uninsurance rate from the covariate set. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Both of our methods find that Medicaid expansion significantly reduced state uninsurance rates by approximately 1.3 to 1.5 percentage points, confirming the first link in the cost-shifting channel.

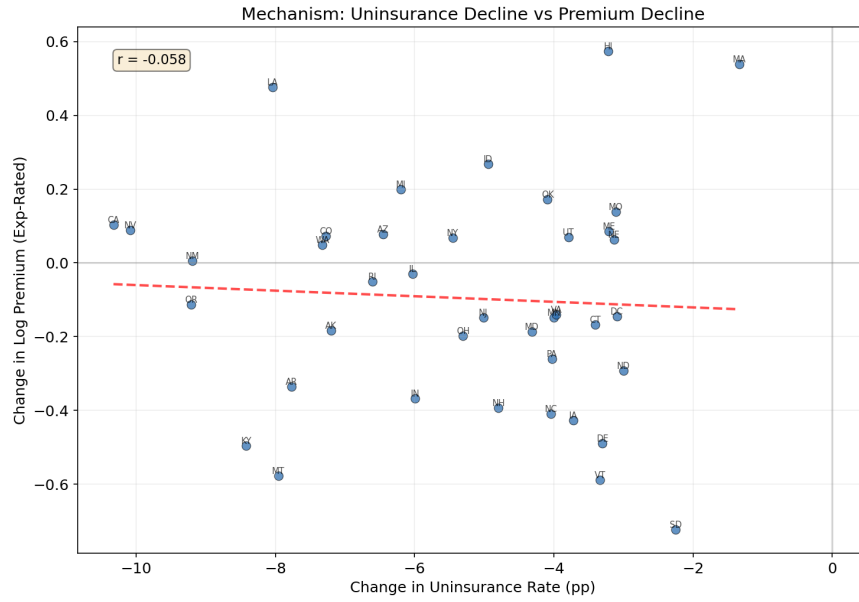


Figure 4. : Mechanism: Change in Uninsurance vs. Change in Log Premium by State

Notes: Each point is an expansion state. The horizontal axis is the difference in mean uninsured rate (post-expansion minus pre-expansion); the vertical axis is the difference in mean log experience-rated premium. The dashed line is a linear fit. States with more data in the pre and post-periods receive equal weight.

Figure 4 plots our cross-state relationship between the change in uninsurance and the change in log experience-rated premiums around expansion. If our proposed cost-shifting channel is the primary mechanism, states with larger uninsurance declines should exhibit larger premium declines. The correlation is near zero ($r = -0.06$), suggesting that while Medicaid expansion significantly reduced state uninsurance rates, the cross-state variation in uninsurance changes does not strongly predict the cross-state variation in premium changes. This weakens our cost-shifting mechanism. State specific factors like baseline premium levels, insurance market concentration, and plan composition likely dominate the cross-state variation which we see in premium changes. This makes it difficult to detect the cost-shifting channel. The DML estimate, which conditions on state-year covariates, is better suited to isolating this channel than the unconditional cross-state correlation.

C. Dose-Response: Pre-Expansion Uninsurance

If the effect on premiums operates through our proposed cost-shifting mechanism, then states with higher pre-expansion uninsurance rates (where more individuals would gain coverage) should show larger premium reductions. We test this by splitting expansion states at the median pre-expansion (2013) uninsurance rates and estimating DML for each of the two groups.

Define the high-dose and low-dose treatment indicators:

$$(6) \quad D_{st}^H = D_{st} \cdot \mathbf{1}[U_s^{\text{pre}} > \tilde{U}]$$

$$(7) \quad D_{st}^L = D_{st} \cdot \mathbf{1}[U_s^{\text{pre}} \leq \tilde{U}]$$

where U_s^{pre} is state s 's 2013 uninsurance rate and $\tilde{U}$ is the median among expansion states. For each dose group, we estimate the PLR model (Equation 3) using only that group's treated states and the never-expanded controls, with Gradient Boosting as the learner.

We find that the effect in high-uninsurance states (-0.269 , $p < 0.001$) is 3.7 times larger than in low-uninsurance states (-0.073 , $p = 0.260$), consistent with our proposed cost-shifting mechanism. The states where Medicaid expansion covered more previously uninsured individuals experienced larger premium reductions.

However, Figure 5 shows that high-uninsurance states also exhibit significant pre-trend violations ($F = 5.86$, $p = 0.003$), while low-uninsurance states have clean pre-trends ($F = 1.27$, $p = 0.315$). This complicates the dose-response interpretation, since the same states driving the larger effect also had pre-existing premium dynamics that differed from controls.

Table 10—: Dose-Response: DML Estimates by Pre-Expansion Uninsurance Level

Subgroup	$\hat{\theta}$	SE	p -value
High pre-expansion uninsurance	−0.269***	0.057	0.000
Low pre-expansion uninsurance	−0.073	0.065	0.260
Pooled (all expansion states)	−0.206***	0.054	0.000

Notes: DML (Gradient Boosting) estimates from Equation (3) with dose-specific treatment indicators. High (low) pre-expansion uninsurance defined as above (at or below) the median 2013 uninsurance rate among expansion states. Each regression includes only the relevant dose group’s treated states and all 10 never-expanded control states. $N = 453$ (high) and $N = 466$ (low). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

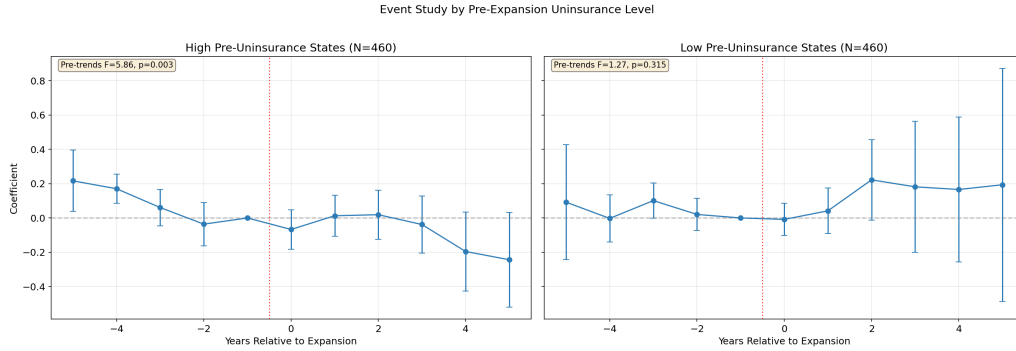


Figure 5. : Event Study by Pre-Expansion Uninsurance Level

Notes: Event study coefficients from Equation (2) estimated separately for high and low pre-expansion uninsurance states. Reference period is $e = -1$. Pre-trends F-test: $F = 5.86$, $p = 0.003$ (high); $F = 1.27$, $p = 0.315$ (low). State and year fixed effects included; standard errors clustered at the state level.

The dose-response results present a mixed picture. The gradient coefficients are directionally consistent with cost-shifting (0.269 and 0.073). States where expansion would cover more uninsured residents show larger premium reductions. The pre-trend analysis complicates this interpretation. Our low-insurance rate subsample where pre-trends are clean ($F = 1.27$, $p = 0.315$) gives an estimate of -0.073 ($p = 0.260$) that is not statistically distinguishable from zero. This estimate may represent a more credible lower bound on the causal effect, identified in the subsample where TWFE and DML assumptions are most defensible. We continue with the question of pre-existing trends in the placebo analysis below.

D. Placebo Test

For our final robustness test we construct a placebo test where we assume Medicaid Expansion occurred in 2008 using only pre-2014 data ($N = 250$, covering 2009–2013). States that eventually expanded receive "treatment" from 2008 onward; never-expansion states remain controls. This placebo effect should be zero.

We estimate the PLR model (Equation 3) with the fake treatment indicator using Gradient Boosting and Lasso (we exclude Random Forest due to overfitting concerns on the small sample). We run the placebo on both experience-rated and non-experience-rated premiums.

Table 11—: Placebo Test: Fake Medicaid Expansion in 2008 (Pre-2014 Data Only)

Outcome	Learner	$\hat{\theta}$	SE	p	Verdict
experience-rated	GBR	-0.173^{***}	0.061	0.004	Fail
	Lasso	-0.272^{***}	0.085	0.001	Fail
non-experience-rated	GBR	0.039	0.025	0.112	Pass
	Lasso	0.114^{***}	0.023	0.000	Fail
<i>For comparison: main results (real expansion, full DML sample)</i>					
experience-rated	GBR	-0.206^{***}	0.054	0.000	—

Notes: Placebo treatment is defined as $\bar{D}_{st} = \mathbf{1}[s \in \text{eventually expanded}] \times \mathbf{1}[t \geq 2008]$, estimated on the pre-2014 sample only ($N = 250$, years 2009–2013). DML uses the same PLR specification and covariate set as the main results. "Pass" indicates $p > 0.10$; 'Fail' indicates $p \leq 0.10$. $***p < 0.01$, $**p < 0.05$, $*p < 0.10$.

The placebo test fails for experience-rated premiums. Both of our learners detect a significant fake effect comparable to our main result estimate (-0.17 to -0.27 vs. -0.21 for the main result). This confirms the pre-trends concern identified in the event study. Medicaid expansion and non-expansion states are already on diverging premium trajectories before any states expanded.

For non-experience-rated premiums, the GBR placebo passes ($p = 0.112$), consistent with the flat pre-trends observed in the event study for that outcome. The Lasso placebo fails, suggesting some sensitivity to learner choice.

This placebo failure does not rule out all of our findings. The placebo is estimated without state fixed effects (the pooled specification), and the failure confirms that between-state variation in experience-rated premiums is contaminated by pre-existing trends. This is precisely why our primary specification includes state dummies. The within-state DML estimates range from 0.081 to 0.217 depending on learner, with the best-fitting model (Gradient Boosting) yielding a marginally significant result; the placebo result reinforces the interpretation that the larger pooled estimates are inflated by pre-existing between-state differences rather than causal effects of expansion.

VII. Limitations

A. Conditional unconfoundedness

For our DML specification conditional unconfoundedness is the main assumption: that after controlling for X_{st} and state fixed effects, treatment assignment is as good as random. This is untestable; we cannot verify that our covariates fully capture all factors that predict both expansion decisions and premium trajectories. There may exist unobserved time-varying state-level characteristics (shifts in regulatory philosophy, insurance market competitiveness, or political environment) that could bias our estimates. The high treatment model R^2 (0.938) indicates that expansion status is nearly fully predicted by covariates and state identity, leaving limited residual variation for identification.

B. Small Control Group

Out of our 51 states tested only ten have not expanded Medicaid. This leaves us with a far smaller control than treatment group. The DML specification model $E[D|X]$ has limited variation to learn from, and inference with few states in the control group is less reliable than if we had a far larger control group.

C. State-year Aggregation

For our specification we aggregate plan-level data from U.S. Government Form 5500 to state-year. This leads to a loss of within-state heterogeneity across plans. This reduces statistical power as plans within the same state may respond differently to expansion depending on various factors like size, industry, or workforce composition, etc. This aggregation is necessary to match the state-level treatment assignments in our specification.

D. Sample Restrictions

Our DML sample is 2009 to 2019 due to covariate availability, we exclude the most recent years due to COVID disruptions and lack of data for 2025. As for earlier years we exclude 1999-2000, and 2004 for data size and validity concerns. This leads to far less years than we potentially could have. The pre-expansion window is five years (2009-2013) and post-expansion coverage varies from one to six years depending on expansion cohort.

VIII. Conclusion

This paper estimates the spillover effects of Medicaid expansion in the United States on employer-sponsored health insurance premiums using Double/Debiased Machine Learning. We utilize plan-level U.S. Department of Labor Form 5500 filings aggregated to a balanced state-year panel of 1,173 observations across 51 states and 23 years (2001 to 2024). We compare our DML estimates to a standard TWFE DiD on the same sample.

Our primary DML specification, which includes state fixed effects to restrict identification to within-state variation, attenuates experience-rated estimates across all learners, with point estimates ranging from 0.081 to 0.217 (8 – –20%). The degree of attenuation varies by learner, ranging from 21% to 62%, indicating sensitivity to how flexibly the nuisance function handles the expanded covariate vector. Removing state fixed effects inflates estimates to 19 – –26%

We document that expansion reduced uninsurance rates, consistent with the first link in the proposed channel: by approximately 1.3–1.5 percentage points. We also show a dose-response gradient that is consistent with cost-shifting: states that have higher pre-expansion uninsurance rates show larger premium effects.

The event study rejects parallel pre-trends for experience-rated premiums, and the placebo test detects significant effects under a fake 2008 expansion, but both diagnostics apply to between-state variation that the primary specification absorbs through state fixed effects. The within-state DML estimate is already modest and insignificant; the robustness tests reinforce the interpretation that larger pooled estimates are inflated by pre-existing between-state differences.

This paper contributes to the existing literature in three main ways. First, this is the first application of DML to this question, and the within-state versus pooled comparison reveals that estimates are highly sensitive to whether state-level heterogeneity is absorbed; a finding with general applicability to DML in policy evaluation settings. Second, the experience-rated versus non-experience-rated distinction provides a built-in falsification test that holds across all specifications: only the outcome that should respond to cost-shifting shows any estimated effect,

with the flexible learners finding null results for non-experience-rated premiums across all specifications. Third, we demonstrate that TWFE’s significant estimate for non-experience-rated premiums is due to confounding that DML eliminates, a practical showcase of ML-based causal methods in applied economics.

For policy, our findings are suggestive but not definitive. The within-state point estimates of 0.081 to 0.217 imply a plausible premium reduction that, if causal, would represent a meaningful benefit beyond the directly eligible population. However, the evidence does not reach statistical significance, and we cannot rule out a zero effect. The ten states that have not expanded Medicaid may be forgoing premium reductions for their employer-sponsored insurance markets, but stronger evidence is needed before firm policy conclusions can be drawn.

Further research can extend this analysis through plan-level DML estimation to exploit within-state heterogeneity, incorporation of post-2019 covariates to extend the sample through the COVID period and beyond.

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